IT Employee Lifecycle Automation System

Automate Onboarding & Offboarding — Zero Manual Steps, Zero Orphaned Accounts

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Project: **IT Employee Lifecycle Automation System**

Demo: **http://localhost:3000** *(run locally)*
Repository: github.com/storrijas97/Automated-User-Onboarding

⚠ The Problem

IT administrators spend hours manually creating Active Directory accounts for new hires — often *after* the employee's first day. Offboarding is even worse: missed notifications leave terminated employees with active credentials, creating security vulnerabilities and compliance risks. Manual provisioning causes inconsistent permissions across employees in identical roles.

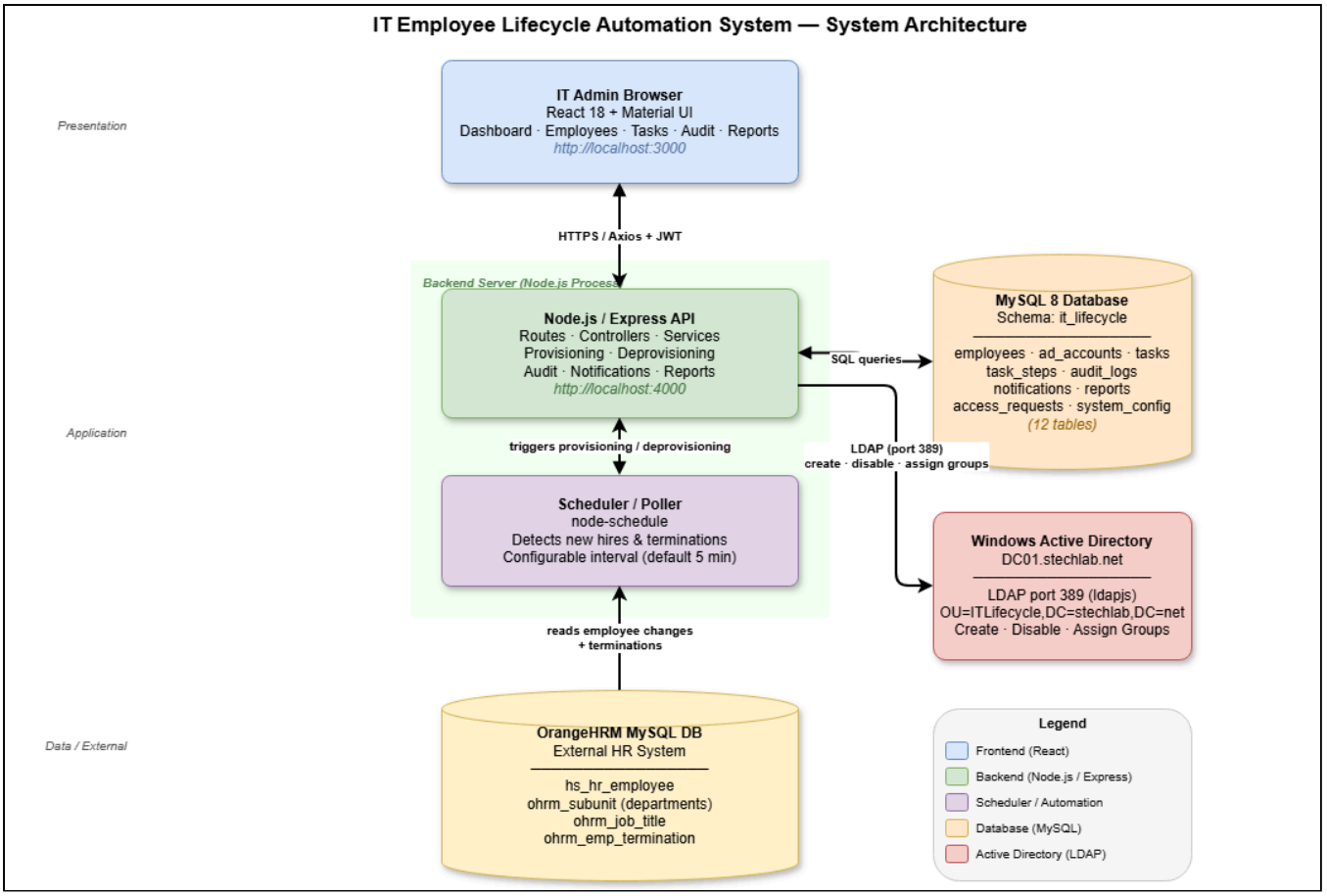
✓ The Solution

This system continuously monitors OrangeHRM for personnel changes and automatically executes the corresponding Active Directory actions. New hires get accounts on Day 1. Departing employees are disabled within minutes of termination. Every action is logged, auditable, and viewable through a clean web dashboard — no tickets, no emails, no manual steps.

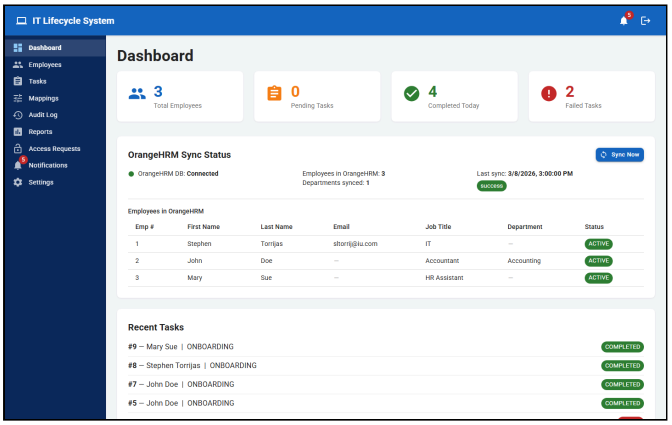
KEY FEATURES

#	Feature	Description
1	OrangeHRM Sync	Polls OrangeHRM on a configurable schedule, keeping employee name, department, and job title current in real time
2	Auto AD Provisioning	Creates Active Directory accounts via LDAP with correct OU placement, display name, email, and enabled status on day one
3	Role-Based Group Assignment	Maps HR department + job title combinations to AD security groups; assignments applied automatically at account creation
4	Auto Deprovisioning	Detects terminations and immediately disables AD accounts, removing all group memberships to eliminate orphaned access
5	Task Management	Tracks every operation (Pending → In Progress → Completed/Failed) with configurable retry logic and priority flagging
6	Audit Log + CSV Export	Full chronological event log with date-range filtering and one-click CSV export for compliance and audit reviews
7	Notification System	Real-time in-app notifications with unread badge counter; can be toggled on/off via system settings
8	Access Request Workflow	Managers submit requests for additional access; IT Admins approve or reject with a full audit trail
9	Compliance Reports	Generate activity summary reports filtered by date range and department; download as CSV
10	System Configuration UI	Adjust polling interval, retry limit, and notification settings at runtime — no code changes required

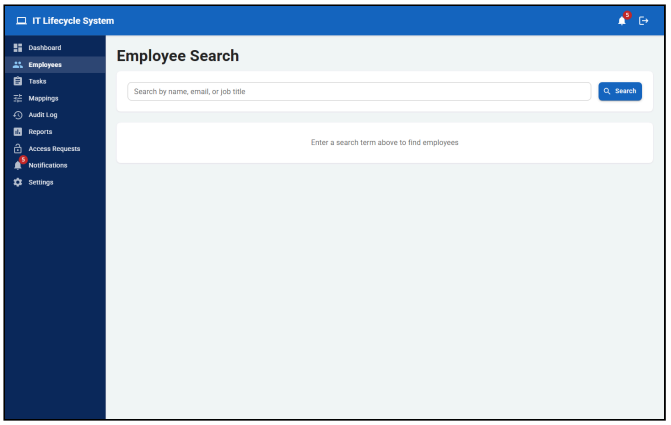
SYSTEM ARCHITECTURE



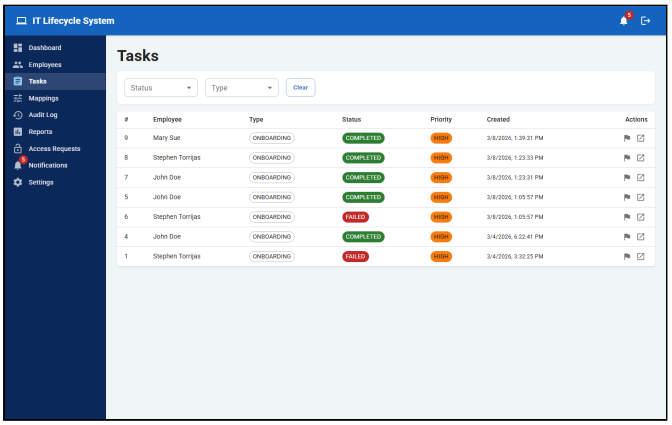
SCREEN SNAPSHOTS



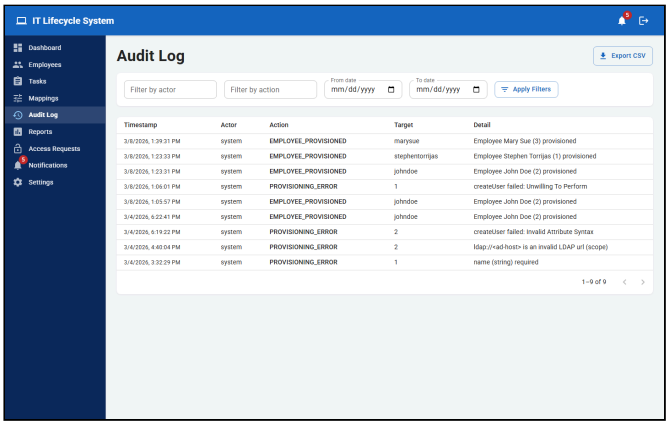
Dashboard — Real-time task statistics & OrangeHRM sync status



Employee Search — Clickable rows with AD account detail modal



Task List — Status filters, priority badges, flag-critical button



Audit Log — Date range filtering & CSV export

SYSTEM REQUIREMENTS

Component	Requirement
Runtime	Node.js 18+ (tested on v25.7.0)
HR System	OrangeHRM 5.x hosted on XAMPP (direct MySQL DB access)
Database	MySQL 8.0+ — schema: it_lifecycle (12 tables)
Directory	Windows Server Active Directory — LDAP port 389
OS	Windows Server 2019/2022 (for Active Directory integration)
Browser	Chrome 120+, Edge 120+, or Firefox 120+
Network	Backend server must reach the AD domain controller and OrangeHRM MySQL host